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Figure 1: Enter Caption

Sherlock Holmes: The Cauliflower Conundrum

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Abstract

This report estimates the probability of harvesting a gold-star cauliflower in the game Stardew Valley and determines the expected number of seeds required to obtain one. The number of seeds planted until success is modeled using a geometric distribution with unknown success probability p , and prior beliefs about p are represented using Beta distributions. Observational data from ten Twitch streams produced 10 gold-star crops across 82 harvest attempts. Bayesian inference is used to combine prior beliefs with observed data to obtain posterior estimates for p . The posterior distribution is derived analytically through Beta–Binomial conjugacy and verified numerically using a Metropolis–Hastings Markov Chain Monte Carlo algorithm. The resulting posterior mean of approximately 0.13 implies an expected 7–8 seeds per successful gold-star harvest. The analysis further illustrates the influence of prior assumptions and confirms that sequential Bayesian updating yields identical posterior results regardless of the order in which data are incorporated.

1 Background and Objective

In Stardew Valley, individuals seek to restore their grandfather’s old farm, shortly after inheriting it. This is achieved through farming, questing, and interacting with the other villagers in the valley. One’s income is dependent solely on the value of the products that they place into their farm’s drop box. One of the most straightforward ways to increase the value of one’s products without simply selling more products is to increase the quality of said goods. Another income source present within the game is the completion of quests. Whether filling the community center quests for improvements around the valley, or simply to gain cold hard cash, Taylor’s goal to ensure that she spends as little as possible on seeds to complete her quest of growing just one gold-star cauliflower is admirable in the economic conditions of today’s day and age.

In this case, as Taylor is trying to ensure that she purchases as few seeds as possible to grow her one gold-star cauliflower, we seek to help Taylor answer several related research questions. First, we determine how many seeds Taylor should expect to purchase in order to successfully grow a gold-star cauliflower. We consider this question both using only the information she obtained from Google and after incorporating the additional data she collected by observing Twitch streams. Second, we investigate how different prior beliefs about the probability of obtaining a gold-star crop might affect the resulting inference, and in particular whether certain prior beliefs could make Taylor’s additional data collection largely unnecessary. Third, we examine how strongly Taylor’s initial Googled belief influences her updated beliefs after observing the Twitch data, and to what extent the prior continues to shape the posterior estimates. Finally, we consider whether the order in which Taylor processes the Twitch data matters. More specifically whether observing one set of streams before another affects the final posterior distribution after updating her beliefs multiple times.

2 Methodology

We contend that X_i , the number of seeds necessary to achieve a gold-star cauliflower with a p chance of success, can be represented by a geometric distribution, while p itself can be represented with a Beta distribution.

$$\begin{aligned} X_i \mid p &\sim \text{Geometric}(p) \\ p &\sim \text{Beta}(\alpha, \beta) \end{aligned}$$

This model is appropriate because each seed represents an independent trial with constant probability of success, and we observe the number of trials until the first success.

Our above random variables have the following PMF/CDFs:

$$\begin{aligned} f(X_i) &= (1 - p)^{X_i - 1} p \\ f(p) &\propto X_i^{\alpha - 1} (1 - X_i)^{\beta - 1} \end{aligned}$$

Under this parameterization, the likelihood for n independent observations is

$$L(p) \propto p^n (1 - p)^{\sum_{i=1}^n X_i - n}.$$

Across the $n = 10$ observed streams, the total number of seeds was $\sum_{i=1}^n X_i = 82$. Thus, the likelihood is proportional to $p^{10} (1 - p)^{72}$.

To explore the impact of prior beliefs, we considered both weak and strong priors. A strong prior represents the case where Taylor’s Googled estimate is treated as highly confident. In this, we are giving it a large “sample size,” the belief is centered around the same success probability but with more pseudo observations. In this scenario, our observed data may have minimal influence on the posterior. This allows us to assess at what point prior beliefs could render data collection effectively moot.

To incorporate prior beliefs, we used the following conjugate Beta priors:

$$\begin{aligned} p &\sim \text{Beta}(2, 8) \\ p &\sim \text{Beta}(20, 80) \\ p &\sim \text{Beta}(3, 17) \\ p &\sim \text{Beta}(1, 1) \\ p &\sim \text{Beta}(2.8, 11.2) \end{aligned}$$

These prior beliefs represent a weak belief in the Googled belief of a 20% success rate, a strong belief in the Googled belief of a 20% success rate, a weak belief in the Googled most commonly occurring success rate of 15%, a uniform belief, and a belief informed by solving for α and β using formulas for expected mean and expected mode, respectively.

For each prior, we computed the posterior distribution and posterior mean of p , and translated this into the expected number of seeds required:

$$E[X] = \frac{1}{p}.$$

We had the option to obtain our Beta(2.8, 11.2) posterior both analytically and numerically. We use the values of median and mode supplied by Taylor and plug them into the expected value and mode equations from a Beta distribution to analytically find our alpha and beta values, which can be found in the respective sections in Analysis and Results. To numerically approximate the posterior distribution of p , we implemented a Metropolis–Hastings (MH) algorithm. Although the posterior distribution can be derived

analytically due to the conjugacy between the Beta prior and the likelihood, the MH algorithm provides a general computational approach for sampling from posterior distributions when closed-form solutions are unavailable or difficult to obtain. Using MH sampling also allows us to verify the analytical result and demonstrate how Markov Chain Monte Carlo methods can approximate the posterior distribution through simulated draws. Because the posterior is proportional to the product of our likelihood and prior, we define the posterior plausibility function:

$$f(p \mid \mathbf{x}_i) \propto L(p)f(p).$$

Starting from an initial value $p_0 = 0.2$, we generated proposal values

$$p' \sim \text{Unif}(p^{(t)} - w, p^{(t)} + w)$$

with half-width $w = 0.05$. Each proposal was accepted with probability

$$\alpha = \min \left(1, \frac{f(p' \mid \mathbf{x}_i)}{f(p^{(t)} \mid \mathbf{x}_i)} \right),$$

such that values with higher posterior density are more likely to be kept while still allowing for some downward moves. Repeating this process generates a Markov chain whose distribution is the posterior distribution of p .

3 Analysis and Results

We address our research questions sequentially. First, we estimate how many cauliflower seeds Taylor should purchase using only the probabilities she obtained from Google. We then incorporate the empirical data gathered from Twitch streams and examine how this information updates her beliefs about the probability of growing a gold-star crop. Finally, we investigate how different prior assumptions influence the posterior distribution and explore when strong prior beliefs could potentially overwhelm the collected data.

Based off of Taylor's Googling, the average chance of a gold-star crop at Farming level 10 with no fertilizer used (the exact same situation she is in) is 20% and the most commonly observed chance is 15%. Given this framework, we argue that strictly based on the Googled information, with $p = 0.2$ and $p = 0.15$ respectively, Taylor should buy 5-7 cauliflower seeds to achieve a gold star cauliflower.

$$\begin{aligned} E[X] &= 1/p \\ 1/0.2 &= 5 \\ 1/0.15 &\approx 6.67 \end{aligned}$$

While these calculations provide an estimate based solely on Taylor's Googled probabilities, they do not incorporate any empirical observations. Taylor also collected data by observing Twitch streams of players farming under similar conditions. We now incorporate this observed data to estimate the probability of obtaining a gold-star cauliflower.

The maximum likelihood estimate from the observed data, with 10 successes out of 82 total attempts is $\hat{p} = \frac{10}{82} \approx 0.122$, representing a 12.2% success rate. If Taylor only had this information, then she should buy 8-9 cauliflower seeds.

$$\frac{1}{n} \sum_{i=1}^n X_i = 8.2$$

To incorporate both Taylor’s prior beliefs and the observed Twitch data, we estimate posterior distributions under several different prior assumptions. These priors range from weakly informative to strongly informative in order to evaluate how much influence prior beliefs may have on the resulting posterior estimates.

Across our priors, our posterior means are represented by the Table 1

Prior	Posterior Mean of p	Expected Seeds ($1/p$)
Beta(1,1)	0.131	7.63
Beta(2,8)	0.130	7.69
Beta(3,17)	0.127	7.87
Beta(20,80)	0.165	6.06

Table 1: Posterior means of p and corresponding expected number of seeds under different prior distribution choices.

Across the weak priors, the posterior mean of p remains close to 0.13, corresponding to an expected 7–8 seeds required to obtain a gold-star cauliflower. Thus, incorporating the Twitch data increases Taylor’s expected number of required seeds relative to the Googled estimate of 5–7 seeds.

We observe that the weak priors (Beta(1,1), Beta(2,8), and Beta(3,17)) produce very similar posterior means (≈ 0.13). The strong prior centered at 0.20 (Beta(20,80)) pulls the posterior upward to 0.165.

Graphically, the weak priors are largely overwhelmed by the likelihood, while the strong prior meaningfully shifts the posterior distribution.

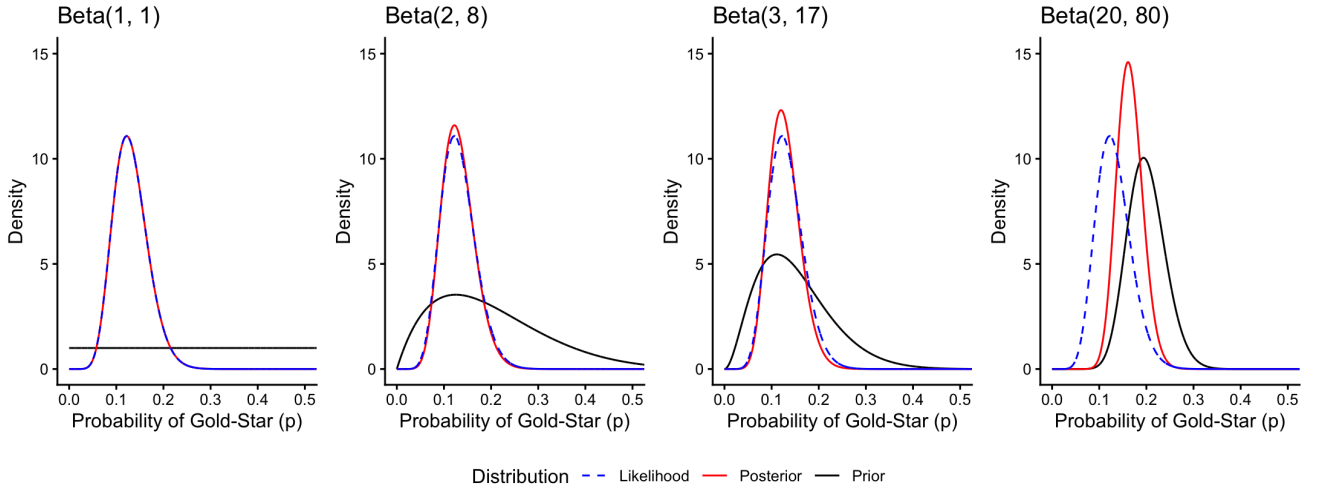


Figure 2: Prior and posterior distributions for four different prior assumptions. Weak priors produce nearly identical posterior distributions, while stronger priors pull the posterior toward the Googled probability of 0.20.

Thus, with only 82 observed crop attempts (10 successes), strongly informative priors, like Beta(20,80), influence the posterior, while weak priors do not.

These results raise an additional question: how strong would Taylor’s prior beliefs need to be in order to meaningfully override the observed data? In other words, under what circumstances might her prior information from Google dominate the posterior distribution regardless of the Twitch observations?

To determine when Taylor’s prior beliefs could render her data collection (and all of the time she spent watching other people farm on Twitch) moot, we compare three models, all centered at 0.2, her googled belief.

As the strength of the prior increases while remaining centered at $p = 0.2$, the posterior distribution becomes increasingly resistant to change from the observed data. The prior $p \sim \text{Beta}(2, 8)$ represents a relatively weak belief in the Googled probability, equivalent to observing roughly 10 prior crops. When combined with the Twitch data (10 successes out of 82 attempts), the posterior shifts noticeably toward the likelihood implied by the data.

In contrast, the prior $p \sim \text{Beta}(20, 80)$ represents a substantially stronger belief, equivalent to approximately 100 prior observations. While the posterior still incorporates the Twitch data, it remains noticeably closer to the prior mean of 0.2 than to the likelihood-based estimate of $\hat{p} = 10/82 \approx 0.122$.

This comparison demonstrates that when prior beliefs are sufficiently strong, additional data may have only a negligible influence on the posterior distribution. In Taylor’s case, her Twitch observations represent 82 crop attempts. Thus, any prior representing several hundred pseudo observations would largely dominate the updating process. Under such a strong prior belief, the posterior distribution remains very close to the prior, indicating that the observed data contribute relatively little information.

Up to this point, we have explored several candidate prior distributions. However, Taylor’s Googled information also provides enough structure to analytically derive a prior distribution consistent with both the reported mean probability and the most commonly observed value. We therefore compute the Beta distribution parameters that correspond to a mean of 0.20 and a mode of 0.15.

Analytically Deriving Posterior Distribution Hyperparameter

Given our mean chance of 0.2 of growing a gold-star cauliflower, and mode of 0.15, we can thus compute beta and alpha for our posterior distribution. For a beta probability distribution function:

$$\begin{aligned} E[X] &= \frac{\alpha}{\alpha + \beta} \\ E[X] &= 0.2 \\ \frac{\alpha}{\alpha + \beta} &= 0.2 \\ \frac{5\alpha}{\alpha + \beta} &= 1 \\ 5\alpha &= \alpha + \beta \\ 4\alpha &= \beta \\ \alpha &= \beta/4 \end{aligned}$$

$$\begin{aligned}
\text{Mode}[X] &= \frac{\alpha - 1}{\alpha + \beta - 2} = 0.15 \\
\frac{20}{3} \cdot \frac{\alpha - 1}{\alpha + \beta - 2} &= 0.15 \cdot \frac{20}{3} \\
\frac{20\alpha - 20}{3\alpha + 3\beta - 6} &= 1 \\
20\alpha - 20 &= 3\alpha + 3\beta - 6 \\
20\alpha &= 3\alpha + 3\beta + 14 \\
17\alpha &= 14 + 3\beta \\
\alpha &= \frac{14 + 3\beta}{17} \\
\text{Thus:} \\
\frac{\beta}{4} &= \frac{14 + 3\beta}{17} \\
17\beta &= 56 + 12\beta \\
5\beta &= 56 \\
\beta &= 56/5 = 11.2 \\
\alpha &= 11.2/4 = 2.8
\end{aligned}$$

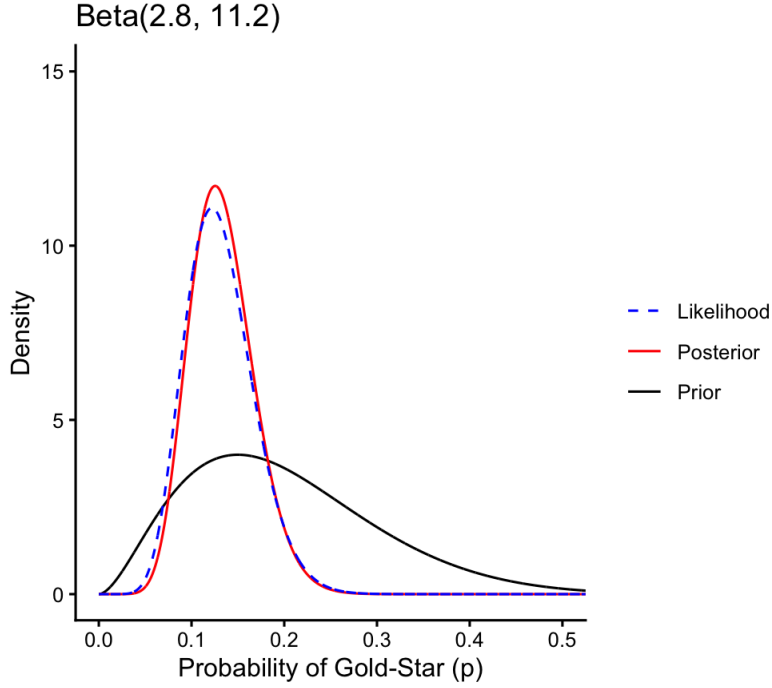


Figure 3: Prior and posterior distributions for the analytically derived prior $p \sim \text{Beta}(2.8, 11.2)$, constructed from the Googled mean probability of 0.20 and mode of 0.15.

Incorporating the Twitch data (10 successes out of 82 attempts) shifts the distribution toward lower values of p , reflecting the lower empirical success rate observed in the streams. The $\text{Beta}(2.8, 11.2)$ prior, along with our Twitch stream data, leaves us with a $\text{Beta}(2.8 + 10, 11.2 + 72) = \text{Beta}(12.8, 83.2)$ posterior distribution, which gives us posterior mean $\frac{\alpha}{\alpha + \beta} = \frac{12.8}{96} \approx 0.133$, which translates to $\frac{1}{0.133} \approx 7.5$ seeds. This gives us a confidence interval of (0.07, 0.20). Given the observed 10 gold-star cauliflowers out of

82 harvest attempts and the prior belief $p \sim \text{Beta}(2.8, 11.2)$ there is a 95% probability that the true probability of harvesting a gold-star cauliflower lies between 0.07 and 0.20, which translates to being 95% confident that we will need between $\frac{1}{0.20} = 5$ and $\frac{1}{0.07} \approx 14$ seeds.

Although the posterior distribution can be derived analytically due to conjugacy between the Beta prior and Binomial likelihood, we also demonstrate a numerical approach using Metropolis–Hastings (MH) sampling. This allows us to approximate the posterior distribution and visually confirm the analytical results.

Numerically Deriving Posterior Distribution Hyperparameter

First, a definition of a function written to return our posterior plausibility, as well as the necessary libraries in R.

Function1

```
library(tidyverse)
library(ggplot2)

post_plaus <- function(p) {
  if(p <= 0 || p >= 1) return(0)
  prior_contrib <- dbeta(p, shape1 = 2.8, shape2 = 11.2)
  like_contrib <- (p^10) * ((1 - p)^72)

  return(prior_contrib * like_contrib)
}
```

And using function 1 to do a Metropolis-Hastings sampling:

Function2

```
p0 <- 0.2
Hastings_p <- rep(NA, 1001)
Hastings_p[1] <- p0

for (i in 1:1000){

  p_prop <- runif(n = 1,
                 min = Hastings_p[i] - 0.05,
                 max = Hastings_p[i] + 0.05)

  Alpha <- min(c(1,
                post_plaus(p_prop) / post_plaus(Hastings_p[i]))))

  Hastings_p[i+1] <- sample(c(p_prop, Hastings_p[i]),
                          size = 1,
                          prob = c(Alpha, 1 - Alpha))
}
```

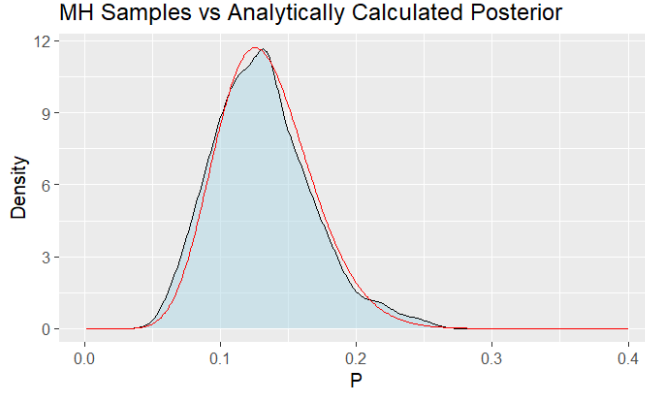


Figure 4: Metropolis–Hastings samples approximating the posterior distribution derived from the prior $p \sim \text{Beta}(2.8, 11.2)$ and the observed Twitch data.

The Metropolis–Hastings samples closely match the analytically derived posterior distribution, confirming that the numerical approximation converges to the same result obtained through the conjugate Beta–Binomial model. We can further confirm these results with our traceplot in figure 5.

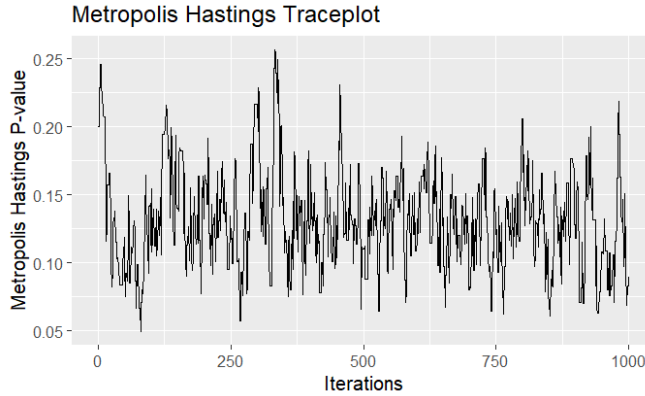


Figure 5: Metropolis Hastings Sampler Traceplot

Our traceplot further supports this conclusion. After the initial iterations, the chain stabilizes and fluctuates around a consistent range of values without any visible upward or downward trend. This behavior suggests that the Markov chain has reached its stationary distribution and is effectively sampling from the posterior distribution of p . The samples move throughout the high-density region of the posterior rather than remaining stuck at particular values, indicating good mixing of the chain. As a result, the simulated draws provide a reliable approximation of the true posterior distribution and allow us to estimate posterior quantities of the mean probability of obtaining a gold-star cauliflower.

Our defined Metropolis-Hastings algorithm provides a valid approximation to the posterior distribution because it is constructed so that the posterior density remains unchanged under the sampling procedure. Recall our acceptance procedure from the methodology section. Our acceptance rule ensures that moves toward regions of higher posterior density are always accepted, while moves toward lower density regions are accepted with a probability proportional to how plausible they are under the posterior. As a result, values with larger posterior density are visited more frequently in the long run.

Finally, looking at how Taylor’s beliefs change or not based on the order she goes about updating them, we note that Bayesian updating with a Beta prior and geometric likelihood depends only on the total number of successes and failures observed in the data. If the prior is $p \sim \text{Beta}(\alpha, \beta)$ and we observe

n successes and f failures, the posterior distribution is

$$p \mid \mathbf{x} \sim \text{Beta}(\alpha + n, \beta + f).$$

Suppose Taylor divides the ten Twitch streams into two groups. Let the first group produce n_1 successes and f_1 failures, and the second group produce n_2 successes and f_2 failures. Updating with the first group and then the second gives

$$\text{Beta}(\alpha + n_1 + n_2, \beta + f_1 + f_2).$$

Reversing the order yields

$$\text{Beta}(\alpha + n_2 + n_1, \beta + f_2 + f_1).$$

As addition is commutative, $n_1 + n_2 = n_2 + n_1$ and $f_1 + f_2 = f_2 + f_1$, meaning the final posterior distribution is identical regardless of the order in which the data are incorporated.

For example, across all ten Twitch streams Taylor observed a total of 10 gold-star crops and 72 non-gold harvests. If the first five streams produced (n_1, f_1) and the remaining five produced (n_2, f_2) such that $n_1 + n_2 = 10$ and $f_1 + f_2 = 72$, sequential updating gives

$$\text{Beta}(\alpha + n_1 + n_2, \beta + f_1 + f_2) = \text{Beta}(\alpha + 10, \beta + 72),$$

regardless of which group is incorporated first. Therefore, it does not matter which set of five streams Taylor observes first as the posterior distribution depends only on the total observed successes and failures across all streams.

4 Discussion

The results suggest that the probability of harvesting a gold-star cauliflower is likely between roughly 7% and 20%, with a posterior mean of approximately 13%. This corresponds to an expected 7–8 seeds required to obtain a gold-quality cauliflower. Incorporating the Twitch stream observations therefore suggests that the true probability may be somewhat lower than the 20% rate Taylor initially found online.

The analysis also demonstrates how prior beliefs can influence posterior inference. When weak priors such as $\text{Beta}(2,8)$ or $\text{Beta}(20,80)$ are used, the posterior estimates are largely determined by the observed data. However, strongly informative priors place greater weight on the prior belief that the success probability is close to 0.20. In this case, the posterior mean increases from roughly 0.13 to about 0.165. This corresponds to a reduction in the expected number of seeds from approximately 7.7 seeds to about 6.1 seeds.

The $\text{Beta}(20,80)$ prior contributes the equivalent of 100 pseudo observations, which is large relative to the available data, the ten Twitch streams that Taylor watched. As a result, the posterior distribution remains closer to the prior belief than to the observed likelihood estimate.

If substantially more Twitch stream data were collected, the influence of the prior distributions would decrease and the posterior estimates would converge toward the likelihood-based estimate. With a larger

sample of observed harvests from Twitch, differences between weak and strong prior assumptions would become less important.

If Taylor treats the googled 20% rate strongly, her estimate of the number of seeds that she needs to buy decreases. If she treats it as weak background information, then the data dominates and suggests that the true probability may be closer to 12-13%.

One limitation of this analysis is the small sample size. Taylor only observed 82 harvest attempts across a limited number of Twitch streams, which introduces considerable uncertainty in the estimated success probability. Additionally, the observed streams may not represent typical gameplay conditions. Differences in farming skill, in-game fertilizer use, or crop quality mechanics could influence the probability of obtaining gold-star crops, meaning the Twitch data may not perfectly reflect Taylor's own gameplay outcomes.

Future analysis could incorporate a larger dataset of observed harvest attempts across more streams or gameplay sessions. Additional modeling could also account for factors such as fertilizer use, farming skill level, or other in-game mechanics that influence crop quality.